

Project Design Phase: Solution Architecture

Date	24 June 2026
Project Name	Licensed Software Installation Requests via ServiceNow
Maximum Marks	4 Marks

1. Architecture Specifications

- **Goal:** To bridge the gap between business needs (manual request bottlenecks) and technical implementation (automated workflow orchestration) using a 3-tier architecture.
- **Structure:**
 - **Tier 1: Presentation Layer:** The ServiceNow Service Portal providing a unified interface for employees.
 - **Tier 2: Business Logic Layer:** ServiceNow Flow Designer and Integration Hub managing approval routing and business logic execution.
 - **Tier 3: Data Layer:** ServiceNow CMDB (Configuration Management Database) and backend tables for request storage and license tracking.
- **Development Phases:**
 - Phase 1: Portal UI Configuration & Catalog Item design.
 - Phase 2: Workflow Automation & Approval Logic development.
 - Phase 3: Integration with Active Directory (User Data) and Asset Management.
 - Phase 4: Testing & Deployment in the production environment.

2. Instructions for Generating Your Architecture Diagram

Since my image generation limit is currently reached, you can easily create the required architecture diagram yourself using tools like Lucidchart, Draw.io, or even PowerPoint. Follow these steps to ensure your diagram is accurate for your project:

1. **Layout:** Create a standard **3-tier diagram**. Draw three horizontal rectangular zones stacked vertically.
2. **Top Tier (Label: "Presentation Layer"):**
 - Place a box inside labeled **"ServiceNow Service Portal"**. Add an icon representing a User/Employee pointing to this box.
3. **Middle Tier (Label: "Application/Logic Layer"):**
 - Place a central process box labeled **"ServiceNow Flow Designer"**.
 - Include smaller boxes connected to it labeled **"Approval Engine"** and **"Integration Hub"**.
4. **Bottom Tier (Label: "Data/Infrastructure Layer"):**
 - Place a database icon labeled **"CMDB / Request Tables"**.

- Place a box labeled "**Software License Repository**".
5. **Data Flow Arrows:**
- Draw an arrow from **User** to the **Portal**.
 - Draw an arrow from **Portal** to **Flow Designer**.
 - Draw an arrow from **Flow Designer** to the **CMDB/License Repository** to represent record updates.
 - Draw a return arrow from **Flow Designer** back to the **Portal** showing the "Status Update" to the user.

References

- <https://c4model.com/>
- <https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/>
- <https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d>

